

Project Report On

**DRIVER DROWSINESS DETECTION THROUGH
FACE MESH SOLUTION USING MACHINE
LEARNING**

Submitted by

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Abstract:

The two main factors that contribute most to accidents on roadways are driver weariness and sleepiness. In recent years, driving when fatigued has become a major factor in vehicle accidents that cause severe injuries, deaths, and monetary losses. Statistics support the need for a reliable system that can identify driver fatigue and alert them in time to avoid an accident. Researchers have looked into three different sorts of signs in an effort to determine whether the driver is sleepy: physiological, behavioral, and vehicle-based markers. In this paper, we presented a system architecture that would alert drivers if they became sleepy while driving up to a certain angle of the driver's head. During driving mode, the driver's level of alertness is also determined. For the purpose of detecting sleepiness, the FaceMesh machine learning solution's range of diverse head position angles has been shown. Additionally, several facial postures are computed to identify driver tiredness with **97.5% accuracy** in the straight face position, and the system reaction time is determined to be about 3 seconds. We have discovered that our suggested method performs better than analogous modern works and is less expensive and hard to create.

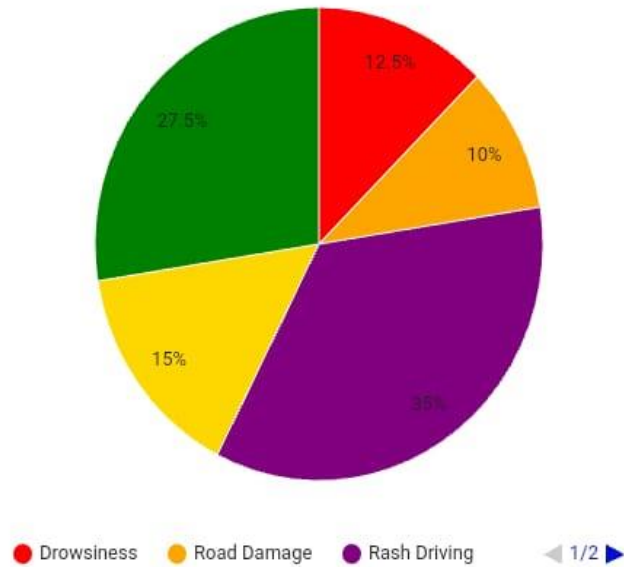
CHAPTER-1

Introduction:

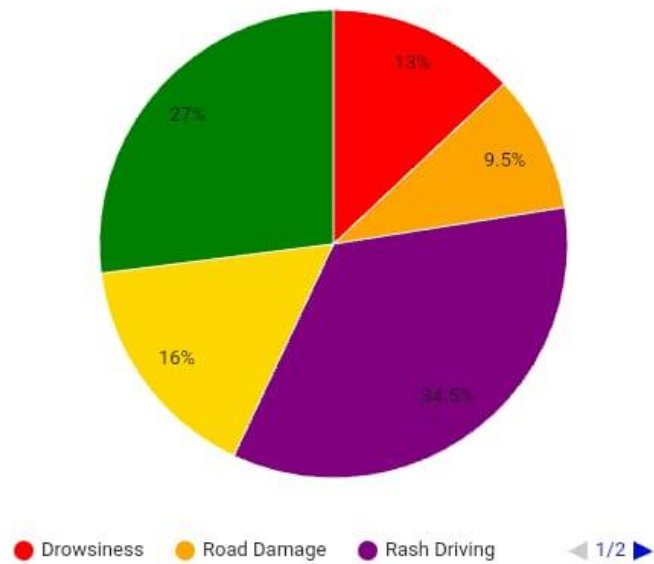
- Drowsiness is estimated to be a factor in 15-25% of all road accidents in India.
- This translates to roughly 22,500 to 37,500 accidents per year, based on the 2022 figure of 1.5 lakh total accidents. (Source: Ministry of Road Transport and Highways, 2022)
- For safe driving, drowsiness detection is required. It is a serious problem that has to be resolved.
- There are four common ways to implement driver drowsiness detection (DDD) systems. Both biological and image based measures can be observed in the drivers themselves. The third measurement, frequently referred to as a vehicle based measurement, is acquired from the actual vehicle. The fourth factor considered is the hybrid measure, which incorporates at least two of the measures previously mentioned.

Pie Charts Based On Road Accidents:

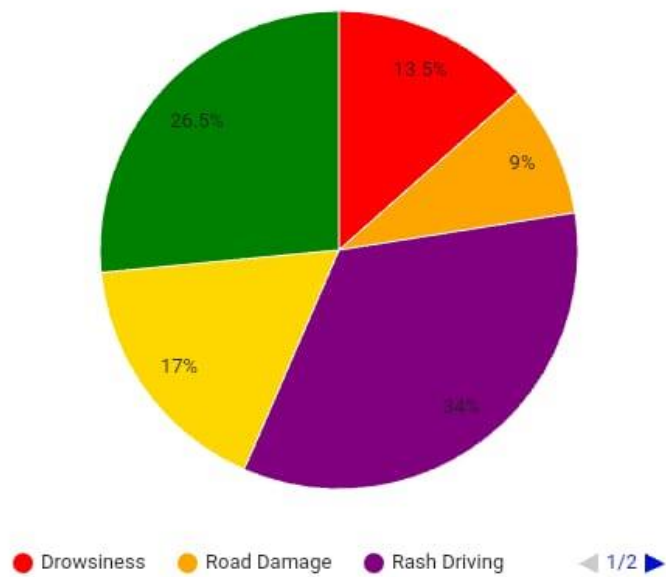
Year 2020 - Accident Causes



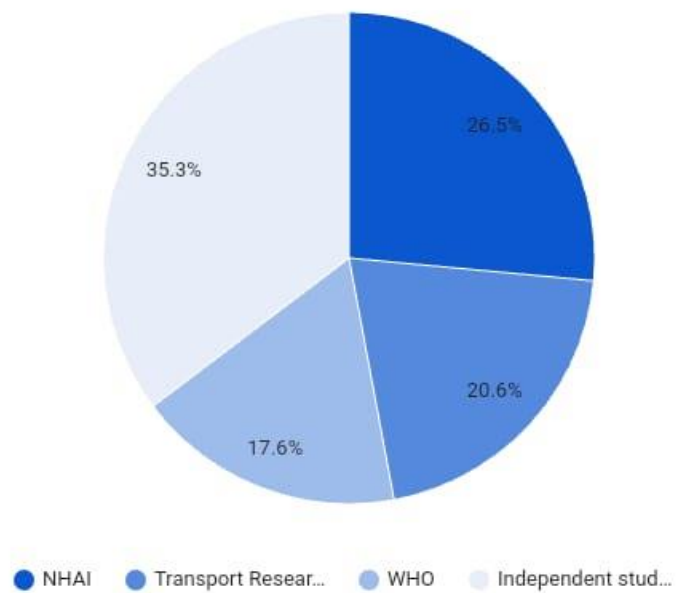
Year 2021 - Accident Causes



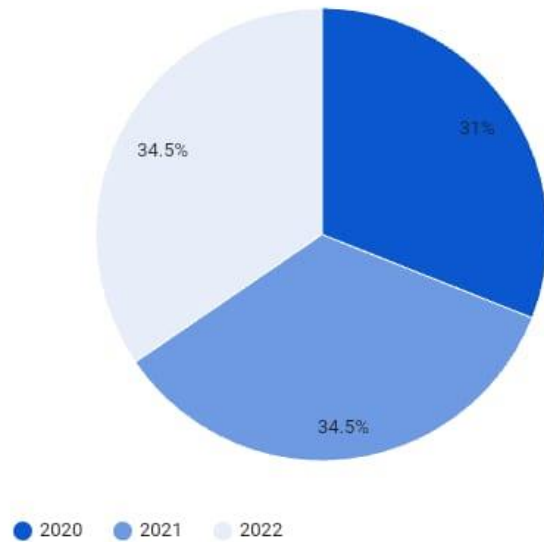
Year 2023 - Accident Causes



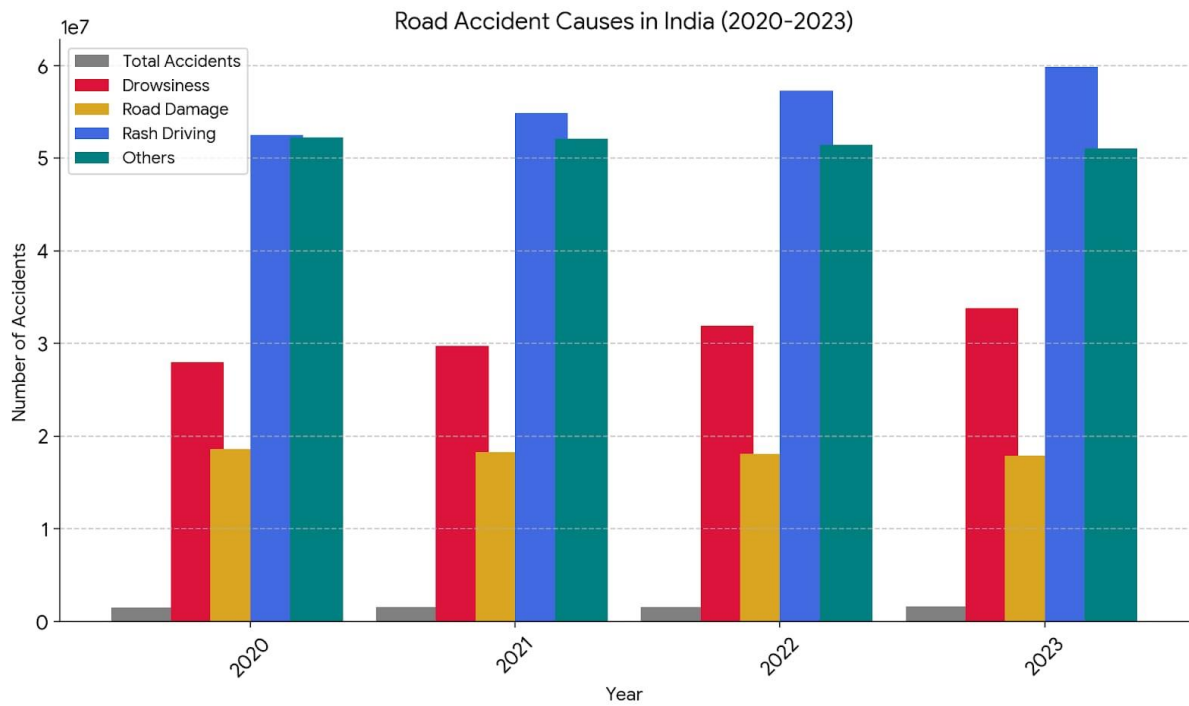
Percentage of Road Accidents Due to Driver Drowsiness in India (Estimates)

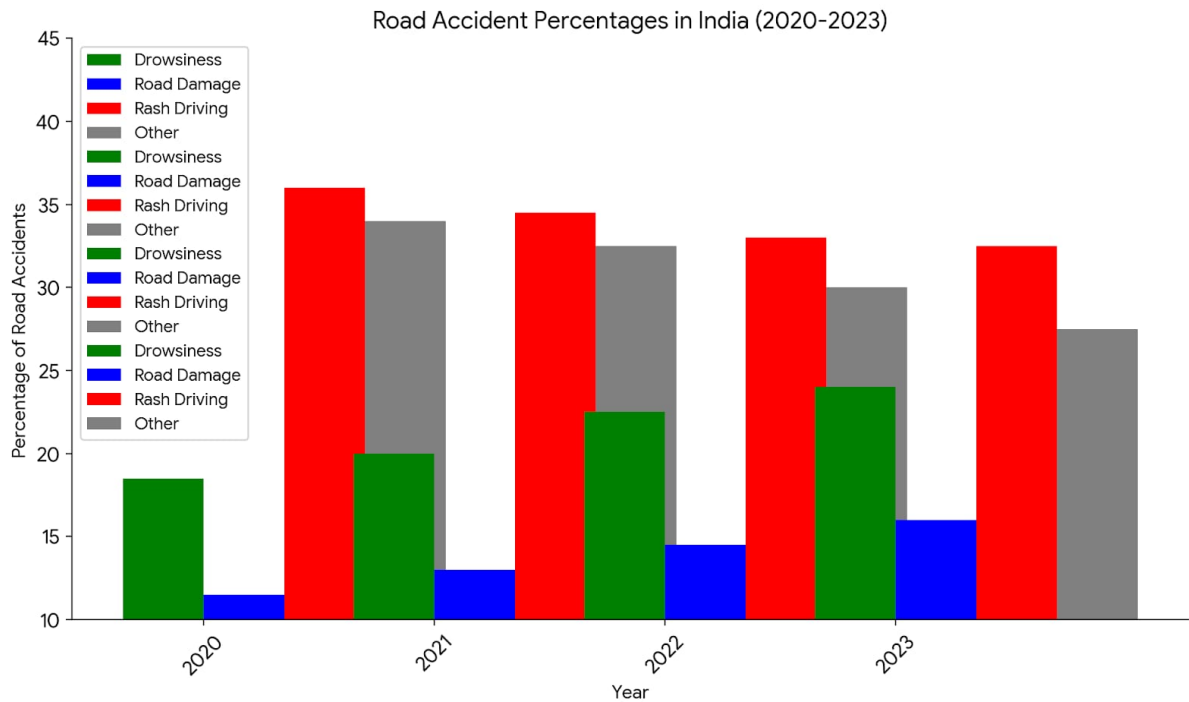


Percentage of Road Accidents due to Driver Drowsiness in India (2020-2023)



Bar Charts based on Road Accidents:





1.1 Problem Statement:

- How can we accurately detect driver drowsiness in real-time, under varying lighting conditions and head poses, using technology and machine learning to prevent accidents caused by driver fatigue?

1.2 Signs and Symptoms of Drowsiness:

Numerous factors, including medications, long workdays, sleep problems, insufficient or poor-quality sleep, and extended periods of awake time, might make you drowsy [19]. Drivers do not suddenly become fatigued without displaying specific indications. These markers include, for instance [9]:

- Difficulty keeping eyes open;
- Yawning;
- Frequent blinking;
- Difficulty concentrating;
- Swerving out of the lane and delayed reaction to traffic;
- Nodding;
- Indefensible differences in speed.

For assessing levels of sleepiness, a precise measurement scale is required. Two commonly used scales are the five-level Wierwille and Ellsworth Drowsiness Scale (WEDS) and the Karolinska Sleepiness Measure (KSS) [20, 21]. According to Shahid et al. [22], KSS is "a scale that analyzes the subjective levels of drowsiness at a certain time of the day." The KSS is a nine-point scale that assesses the verbal complaints of tiredness made by drivers [23]. The nine KSS scores are summarized in Table 1. The WEDS results for the five levels are listed in Table 2. According to Saito et al. [24], level one can be distinguished by quick eye movement and steady blink rates. At stage two, there is slow eye movement. The driver can touch his face, yawn, and blink slowly at level three. In stage four, the driver is seen moving pointlessly, yawning constantly, blinking

excessively, and inhaling deeply. The eyes are nearly closed and the head nods at the sixth level. This scale is also widely used because these levels are determined by assessing the driver's facial expressions.

Table 1. Karolinska sleepiness scale (KSS)

Scale	Verbal Description
1	Extremely alert
2	Very alert
3	Alert
4	Fairly alert
5	Neither attentive nor drowsy
6	Some symbols of drowsiness
7	Sleepy, but making little effort to stay awake
8	Sleepy, but making effort to stay awake
9	Very sleepy, great exertion to retain alert

Table 2. Wierwille and Ellsworth drowsiness scale (WEDS)

Levels	Verbal Description
1	Not drowsy
2	Slightly drowsy
3	Discreetly drowsy
4	Meaningfully drowsy
5	Very drowsy

1.3 Drowsiness Detection Measures

There are four common ways to implement driver drowsiness detection (DDD) systems. Both biological and image-based measures can be observed in the drivers themselves. The third measurement, frequently referred to as a vehicle-based measurement, is acquired from the actual vehicle. The fourth factor considered is the hybrid measure, which incorporates at least two of the measures previously mentioned.

(i) Image-based Measures:

- Eye-based method
- Mouth-based method
- Head-based method

(ii) Biological-based Measures:

- Brain signal-based method
- Respiratory signal-based method
- Heart signal-based method
- Skin signal-based method
- Eye signal-based method
- Muscles signal-based method

(iii) Vehicle-based Measures:

- Steering wheel-based method
- Lane deviation-based method

(iv) Hybrid-based Measures:

- Vehicle and Image-based method
- Biological and Image-based method
- Vehicle and Biological-based method

1.4 Methodology:

1.4.1 Face and Feature Detection:

- Camera setup
- Face detection
- Feature extraction

1.4.2 Head Position Tracking:

- Angle estimation: Algorithms calculate angles such as:
 - Pitch: Vertical tilt (up/down)
 - Yaw: Rotation side-to-side
 - Roll: Rotation towards shoulders
- Distance tracking

1.4.3 Drowsiness Indicators:

- Nodding
- Extended downward tilt
- Slow, swaying movements

1.4.4 Alerting System:

- Thresholds and decision-making
- Alerts: Visual, auditory, or haptic alerts are triggered to warn the driver, such as: Audible beeps or alarms.

1.5 Existing System:

Existing driver drowsiness detection systems use various techniques such as monitoring the driver's eye blink rate, yawning, heart rate, and steering wheel technique. These systems aim to detect drowsiness and alert the driver to prevent accidents caused by fatigue. The effectiveness of these systems varies, with some achieving high accuracy rates. However, they also have limitations, such as being affected by external factors like lighting conditions and driver behavior.

1.6 Proposed System:

The proposed system uses real-time facial data captured by a camera and processed by the Facemesh algorithm to detect drowsiness events. The system also includes a raspberry pi, and a sound box to alert the driver in case of drowsiness. The proposed system has the potential to improve driver safety and reduce the risk of accidents caused by drowsy driving.

1.7 Design Process:

1.7.1 Hardware Arrangement:

To implement the device, we use some electronics components. All of them discussed below.

1. Raspberry Pi 4.0
2. Webcam
3. Speaker
4. HDMI cable/USB cable
5. Power cable

6. Wire cables

1. **Raspberry Pi 4.0:** The Raspberry Pi is a low-cost Linux computer with a set of GPIO (general purpose input/output) ports for controlling electronics with the Internet of Things (IoT).
2. **Webcam:** A webcam is a digital video camera in a personal computer. One of its primary uses is to share images online. It used in both image capture and instant messaging.
3. **Speaker:** How high or low a sound is in pitch is determined by its frequency. It produces high-frequency sound waves, while the low-frequency sounds of a bass guitar or kick drum are produced.
4. **HDMI cable/USB cable:** High-Definition Multimedia Interface at a very high resolution. It's the most used format for sending high-definition video and audio via a single connection from one device to another.
5. **Power cable:** A power cable is a type of electrical cable that consists of one or more electrical conductors.
6. **Wire cables:** Wires are singular conductive elements, these conductors are typically composed of copper or aluminium. Usually, the wires are exposed and coiled. However, some of the wires have a thin PVC coating.

1.7.2 Software:

To run the setup hardware, we use an operating system and a programming language. We use:

- **Python:** New data-handling classes, faster asynchronous I/O, and improvements to script compilation and garbage collection may all be found in Python 3.7. Python 3.7, the newest version of the language, has been released to the public with the goal of simplifying previously intractable problems.
- Operating system used: **linux**
- ✓ **LINUX:** Linux is perfect for use with any kind of server application because of its support for high-volume and multithreaded workloads.
- Operating system designed for use on a desktop computer.

CHAPTER – 2

LITERATURE SURVEY:

1.Real-Time Driver Drowsiness Detection System Using Facial Features

Author: *Wanghua deng and Ruoxue wu*

The face is an important part of the body, conveys a lot of information. Imagine a system that watches your face while you drive and can tell if you're getting tired. DriCare does exactly that! It uses a camera to analyze your blinking, yawning, and how long your eyes stay closed to detect fatigue. No awkward headbands or sensors needed! DriCare even improved its accuracy by tracking your face better and understanding specific features like your eyes and mouth. The result? A 92% chance of catching drowsy drivers before they make a mistake. This could be a game-changer for road safety

2.Early Identification and Detection of Driver Drowsiness by Hybrid Machine Learning

Author: *Ayman Altameem, Ankit Kumar and Sandeep Kumar*

Drunkenness or exhaustion is a leading cause of car accidents, with severe implications for road safety. More fatal accidents could be avoided if fatigued drivers were warned ahead of time. Several drowsiness detection technologies to monitor for signs of inattention while driving and notifying the driver can be adopted. Sensors in self-driving cars must detect if a driver is sleepy, angry, or experiencing extreme changes in their emotions, such as anger. These sensors must constantly monitor the drivers facial expressions and detect facial landmarks in order to extract the drivers state of expression presentation and determine whether they are driving safely. As soon as the system detects such changes, it takes control of the vehicle, immediately slows it down, and alerts the driver by sounding an alarm to make them aware of the situation. The proposed system will be integrated with the vehicles electronics, tracking the vehicles statistics and providing more accurate results.

3. Smart Edge-Based Driver Drowsiness Detection in Mobile Crowdsourcing

Author: *Hanane Lamaazi and Ruba Ali Fadul*

This proposes a solution to prevent accidents caused by drowsy drivers using smart edge computing. The solution uses mobile devices to analyze the driver's condition without sharing their data, and a distributed edge architecture to manage the area of interest. The system confirms drowsiness when the driver's status matches the observed car path. The proposed approach uses facial expressions and acceleration readings to detect drowsiness and relies on a two-layer LSTM algorithm for detection.

4. Privacy-Preserving Federated Transfer Learning for Driver Drowsiness Detection

Author: *LinLin Zhang and Hideo Saito*

It proposes a method for detecting driver drowsiness while preserving privacy and reducing communication costs in the federated learning system. The proposed method uses fine-tuning transfer learning and a CKKS-based privacy-preserving protocol to improve model performance and protect personal data. The method is evaluated on two datasets, and the experimental results show that it outperforms other methods in terms of accuracy and communication efficiency. This concludes that the proposed method can contribute to improving road safety and preventing traffic accidents caused by driver drowsiness.

5. Real-Time Driver Drowsiness Detection using Computer Vision

Author: *Mahek Jain and Bhavya Bhagerathi*

In this method the system continuously monitors the driver's face, particularly the eyes and mouth, and alerts the driver through voice warnings upon detecting signs of drowsiness or yawning. The proposed technology has been tested using the iBUG-300w Dataset and has shown promising results in detecting drowsiness and yawning under various conditions. This system has the potential to enhance transportation safety and prevent road accidents.

6. Driver Drowsiness Detection System – An Approach By Machine Learning Application

Author: *Jagbeer Singh, Ritika Kanojia and Rishika Singh*

In this we addresses the critical issue of drowsy driving, which contributes significantly to road accidents and fatalities worldwide. By leveraging artificial intelligence algorithms, the research

focuses on real-time detection of driver drowsiness through facial and eye tracking. The system matches extracted eye images with a dataset to determine if the driver's eyes are closed for an extended period, triggering an alarm to alert the driver. The ultimate aim is to achieve an 80% accuracy in drowsiness detection, thereby contributing to the reduction of road accidents.

7. Driver Drowsiness Recognition Based on Computer Vision Technology

Author: *Wei Zhang and Bo Cheng*

It introduces a non-intrusive method for detecting driver drowsiness using eye-tracking and image processing. By employing a robust eye detection algorithm, the system addresses challenges related to varying illumination and driver posture. Six measures, including eyelid closure percentage, blink frequency, and eye opening/closing velocities, are calculated and combined using Fisher's linear discriminant functions to create an independent index. Driving simulator experiments with six participants validate the feasibility of this video-based drowsiness recognition method, achieving an accuracy of 86%.

8. IoT-Based Non-Intrusive Automated Driver Drowsiness Monitoring Framework for Logistics and Public Transport Applications to Enhance Road Safety

Author: *M. Adil khan and Tahir Nawaz*

It represents an end-to-end non-intrusive IoT-based framework for continuous monitoring of driver behaviours, tailored for logistic and public transport applications. The framework incorporates an embedded system, edge computing and cloud computing modules, and a mobile phone application to provide a comprehensive solution for drowsiness detection, monitoring, and driver evaluation. Drowsiness detection is achieved through image processing-based techniques to identify sleeping, yawning, and distraction behaviours. Extensive experimental testing demonstrates an overall accuracy of 96% and highlights the enhanced robustness, portability, and usability of the proposed framework.

9. Heart Rate Variability-Based Driver Drowsiness Detection and Its Validation With EEG

Author: *Koichi Fujiwara, Keisuke Kamata and Chikao Nakayama*

This work introduces a driver drowsiness detection algorithm based on heart rate variability prevent fatal car accidents caused by drowsy driving. By monitoring eight HRV features using multivariate statistical process control, changes in HRV indicative of drowsiness are detected. The proposed (HRV) analysis, performance was evaluated through an experiment using a driving simulator, demonstrating its effectiveness in detecting drowsiness. The study also validates the proposed method by comparing it with electroencephalography (EEG)-based sleep scoring, showing promising results. This HRV-based anomaly detection framework has the potential to contribute significantly to preventing accidents caused by drowsy driving.

10. Sensor Applications and Physiological Features in Drivers' Drowsiness Detection: A Review

Author: *Anuva Chowdhury , Rajan Shankaran and Manolya Kavakli*

Drowsiness in drivers is a serious concern due to the high number of fatalities on the road each year. This paper reviews the development of novel and effective methods for detecting driver drowsiness, including vehicle-based, behavioural, and physiological methods. Physiological signals are more accurate in defining the transition between wakefulness and sleep, and the proposed methods measure these signals using various sensors to monitor the driver's physiological parameters continually. This paper provides an insightful review of key physiological parameters related to drowsiness, their advantages, and limitations. The aim is to identify, describe, and analyze these parameters to improve driver safety and prevent accidents caused by drowsy driving.

11. Cross-Subject Zero Calibration Driver's Drowsiness Detection: Exploring Spatiotemporal Image Encoding of EEG Signals for Convolutional Neural Network Classification

Author: *Joao Ruivo Paulo, Gabriel Pires, and Urbano J. Nunes*

By using EEG signals during sustained-attention driving, with a focus on cross-subject zero calibration. Highlighting the significance of detecting drowsiness to prevent accidents, the authors propose utilizing spatiotemporal image encoding of EEG signals for deep convolutional neural network (CNN) classification. Their comparison of two techniques using a dataset of 27 subjects demonstrates a superior balanced accuracy of up to 75.87% for leave-

one-out cross-validation, showcasing the potential for pursuing cross-subject zero calibration design.

12. Driver Drowsiness Classification Using Fuzzy Wavelet-Packet-Based Feature-Extraction Algorithm

Author: *Rami N. Khushaba , Sarath Kodagoda, Sara Lal*

In this we addresses the critical issue of driver drowsiness and its impact on road safety. The authors aim to enhance drowsiness detection by extracting information from physiological signals, including EEG, EOG, and ECG, during a simulated driving test. They propose an efficient feature-extraction method, Fuzzy Mutual Information-based Wavelet Packet Transform (FMIWPT), to classify driver drowsiness levels. The method utilizes fuzzy memberships to estimate mutual information, resulting in accurate information content measurement. Evaluation on datasets from 31 drivers demonstrates the effectiveness of FMIWPT, achieving an average classification accuracy of 95%–97% across all subjects.

13. Driver Drowsiness Detection using Python

Author: *Dr .T. Dinesh Kumar, M. Chandrahas Reddy, N. Chandra Sai Raghav*

It aims to develop a driver drowsiness detection system for the Intelligent Transportation System to prevent accidents caused by driver errors. The proposed system utilizes computer vision techniques to monitor blink patterns and eye movements non-intrusively. The simulation results demonstrate the system's ability to detect drowsiness despite the driver wearing spectacles and the amount of darkness inside the vehicle, with a detection time of less than two seconds. The system alerts the driver in real-time through alarms. The accuracy of drowsiness detection is predicted using a Convolution Neural Network (CNN) technique, which can benefit image processing, picture analysis, image pattern recognition, and human vision.

14. Driver Drowsiness Detection using Machine Learning Approach

Author: *Priyanka Basavaraj Murdeshwar, Shruthi Tharanath Salian, Sharath D S*

It presents a machine learning approach for detecting driver drowsiness, which is a significant cause of road accidents. The proposed approach involves face detection to locate the driver's

eyes, which are then tracked in subsequent frames for drowsiness detection. The system uses image processing to recognize the driver's face and extract the eye images for drowsiness detection using the Eye Aspect Ratio (EAR). The proposed system was tested on a Raspberry Pi 3 Model B with 1GB RAM and achieved an average correct rate of 95.0% for eye location and tracking. The proposed approach is a low-cost and effective solution for real-time driver drowsiness detection.

15. Real-Time System for Driver Fatigue Detection Based on a Recurrent Neuronal Network

Author: *Younes Ed-Doughmi, Najlae Idrissi*

It addresses the global concern of road safety and the increasing number of car accident fatalities. The analysis of driver behaviour, particularly drowsiness, has become a leading research subject in recent years. This paper proposes a method for analyzing and predicting driver drowsiness using a Recurrent Neural Network applied over a sequence frame of the driver's face. The proposed method utilizes a dataset to shape and validate the model and implements a multi-layer model-based 3D Convolutional Networks architecture to detect driver drowsiness. After a training session, the proposed method achieved a promising accuracy of 92%, making it possible to develop a real-time driver monitoring system to reduce road accidents.

16. Drowsiness Detection System in Real Time Based on Behavioural Characteristics of Driver using Machine Learning Approach

Author: *Gauri Adarsh, Vineet Singh*

It aims to develop a reliable and efficient system for real-time detection of drowsiness using OpenCV, D lib, and facial landmark detection technologies. The proposed system utilizes a 68 facial landmark detector to assess whether the driver's eyes are closed or open and analyzes the data collected using machine learning methods to detect patterns associated with drowsiness. When drowsiness is detected, the system incorporates a warning mechanism, such as an alarm or a vibration in the steering wheel, to notify the driver. The system was evaluated with different drivers and driving conditions and achieved an average accuracy rate of 94% for identifying tiredness in drivers. The proposed method is less intrusive and more

economical than conventional sleepiness detection techniques and can assist in preventing drowsy driving incidents, enhancing road safety, and saving lives.

17. Real-Time Driver Drowsiness Detection and Classification on Embedded Systems Using Machine Learning Algorithms

Author: *Muhammet Emin Sahin*

This study introduces a real-time driver drowsiness detection and classification system implemented on Jetson Nano and Jetson TX2 embedded systems using machine learning algorithms. The system employs feature extraction, normalization, and classification processes, with the Multi-Layer Perceptron classifier demonstrating superior performance. The study's findings highlight the high accuracy of this hardware-based system in detecting drowsy driving and its potential for in-vehicle use, emphasizing its critical role in enhancing road safety. The achieved accuracy, F1 score, and AUC underscore the effectiveness of the proposed system in early detection and identification of a driver's drowsy state, addressing a significant concern in traffic safety.

18. Driver Drowsiness Detection

Author: *Balaaji.S1, Sri hari, Dr R. P. S. Manikandan1*

It represents a driver drowsiness detection architecture that utilizes deep learning CNN framework and object detection training to detect somnolence in auto mobilists. The system comprises two phases, image data acquisition, and deep learning algorithm classification. The hardware design includes Raspberry pi, camera, GSM module, and a speaker, with the entire training AI software uploaded to the standalone module. Real-time image acquisition and forecasting of driver drowsiness detection demonstrate a delicacy of 95 to 96, with the Re LU activation function improving computational effectiveness. The system's accuracy and real-time status output to warn the auto mobilist during sleepiness state highlight its potential to enhance driver safety. The study's findings emphasize the importance of Artificial Intelligence and Machine learning approaches in advancing driver safety technology

19. Real-Time Driver Drowsiness Detection Using YOLO and Machine Learning Technique

Author: *Chanda Manideep, D L N .N .Manikant, Dr V Vaissnave*

This study introduces a system that utilizes the YOLO algorithm to assess a driver's image and detect signs of tiredness in real-time. By analysing facial movements, including blinking,

yawning, and eye closure time, the system identifies sleepiness with high accuracy. It provides a visual and auditory warning if the driver is nodding off, aiming to enhance transportation safety by reducing accidents caused by drowsy driving. The system's novel approach, combining eye aspect ratio with eye points, and utilizing a face-tracking algorithm and 68 key points for facial area detection, contributes to its high accuracy in identifying driver fatigue. The study's experiments demonstrate the system's effectiveness, positioning it as a practical option for detecting drowsiness in motorists.

20. REAL TIME SLEEP/DROWSINESS DETECTION

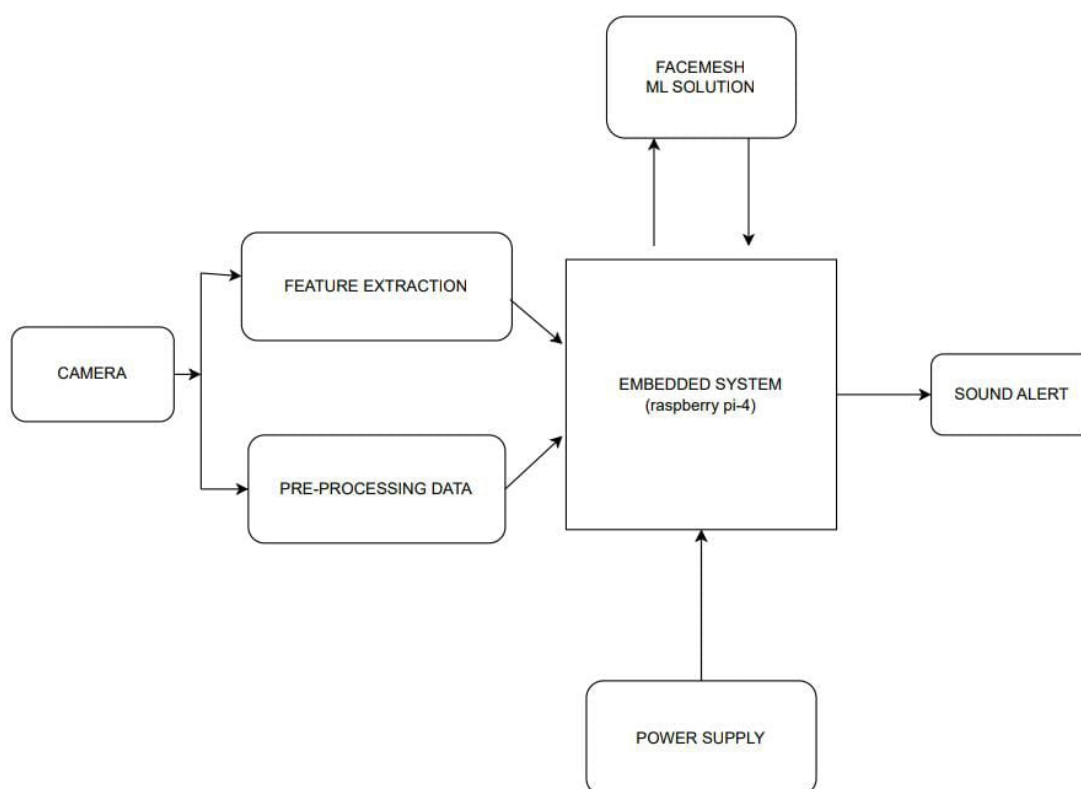
Author: *Dr Varsha Shah*

It aims to develop a nonintrusive system that can detect driver fatigue and issue a timely warning to prevent accidents caused by drowsy driving. By monitoring the driver's eyes using a camera and developing an algorithm, the system can detect symptoms of fatigue early enough to avoid the person from sleeping. The warning will be manually deactivated through a dialog containing simple mathematical operations that must be answered correctly. If the driver feels drowsy, a warning signal in the form of text and sound will be given. The system's ability to detect drowsiness/fatigue can be used as a record of driver performance.

CHAPTER-2

SYSTEM DESIGN

2.1 Block Diagram:



2. Users:

The users for a driver drowsiness detection system can vary based on the specific application and context. Here are some potential users for a driver drowsiness detection system:

1. **Individual Drivers:** Individual drivers, including professional drivers, and truck drivers, can benefit from using drowsiness detection systems to monitor their own alertness and prevent accidents caused by drowsy driving.
2. **Automotive Manufacturers:** Car manufacturers can integrate drowsiness detection systems into their vehicles as a safety feature to provide real-time alerts to drivers when signs of drowsiness are detected, preventing accidents.
3. **Transportation Authorities:** Public transportation authorities, such as bus and train operators, can utilize drowsiness detection systems to increase the safety of passengers and prevent accidents caused by drowsy drivers.
4. **Research and Development Organizations:** Research labs, and organizations focused on transportation safety can use drowsiness detection systems for studying driver behaviour and developing advanced safety technologies.

3. Modules:

- The modules used in this system are listed below:

1. Webcam Module
2. Video Capture
3. Face Detection & Feature Extraction
4. Face Mesh Algorithm Module

- 5. Raspberry Pi Module
- 6. Machine Learning Model
- 7. Alerting Module

1. Webcam Module: The webcam is used to capture real-time face data, which is processed by the Face Mesh Algorithm Module.

2. Video Capture: Access a video stream from a camera, the driver-facing camera within the vehicle. Captures real-time video frames for processing.

3. Face Detection & Feature Extraction:

- Applies a face mesh model to the detected face region. Generates a detailed 3D representation of the face using a network of key points (landmarks).
- Extracts relevant features from the face mesh to detect drowsiness.
Key features include: Head pose estimation to detect nodding

4. Face Mesh Algorithm Module: This module processes real-time face data using the Media Pipe Face Mesh machine learning function, which provides 468 facial landmarks on the face.

5. Raspberry Pi Module: The Raspberry Pi is programmed with Face Mesh and connected with the camera, and alarm by connecting wires. It serves as a main component in the system.

6. Machine Learning Model: Receives extracted features as input. Classifies the driver's state as drowsy or alert.

- Common model types: Support vector machines (SVMs), Artificial neural networks (ANNs), Convolutional neural networks (CNNs).

7. Alerting Module: If drowsiness is detected, triggers an alert to the driver. Common alert modalities: Audible alarms.

4. UML Diagrams

Unified modelling language (UML), is a standardized general-purpose visual modelling language in the field of Software Engineering. It is used for specifying, visualizing, constructing, and documenting the primary artifacts of the software system. It helps in designing and characterizing, especially those software systems that incorporate the concept of Object orientation. It describes the working of both the software and hardware systems.

Characteristics of UML:

The UML has the following features:

1. It is a generalized modelling language.
2. It is distinct from other programming languages like C++, Python, etc.
3. It is interrelated to object-oriented analysis and design.
4. It is used to visualize the workflow of the system.
5. It is a pictorial language, used to generate powerful modelling artifacts.

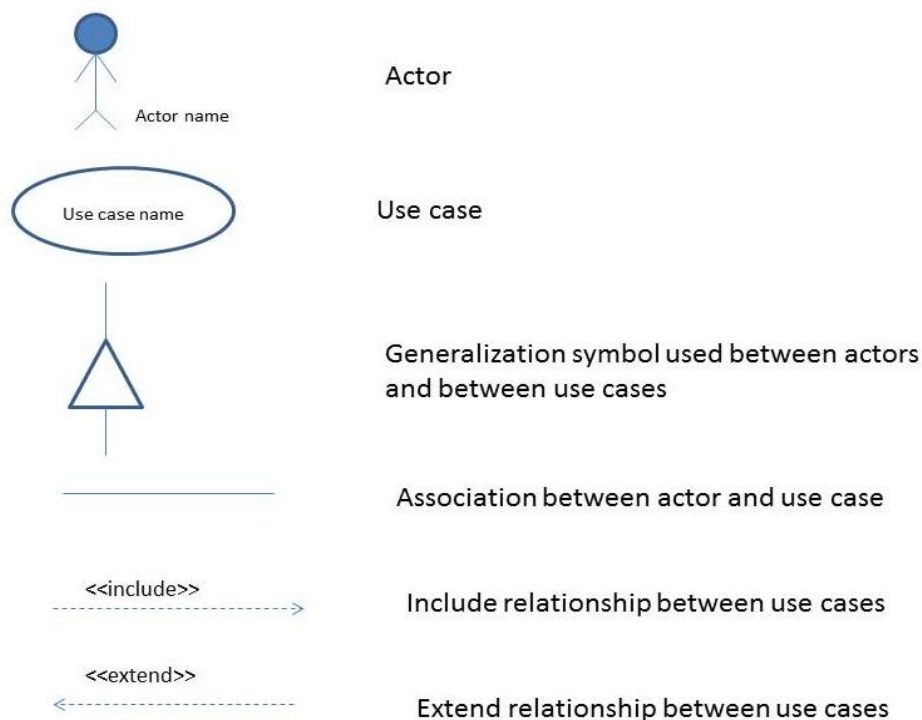
1. Use Case Diagram

A Use Case Diagram is a vital tool in system design, it provides a visual representation of how users interact with a system. It serves as a blueprint for understanding the functional requirements of a system from a user's perspective, aiding in the communication between stakeholders and guiding the development process.

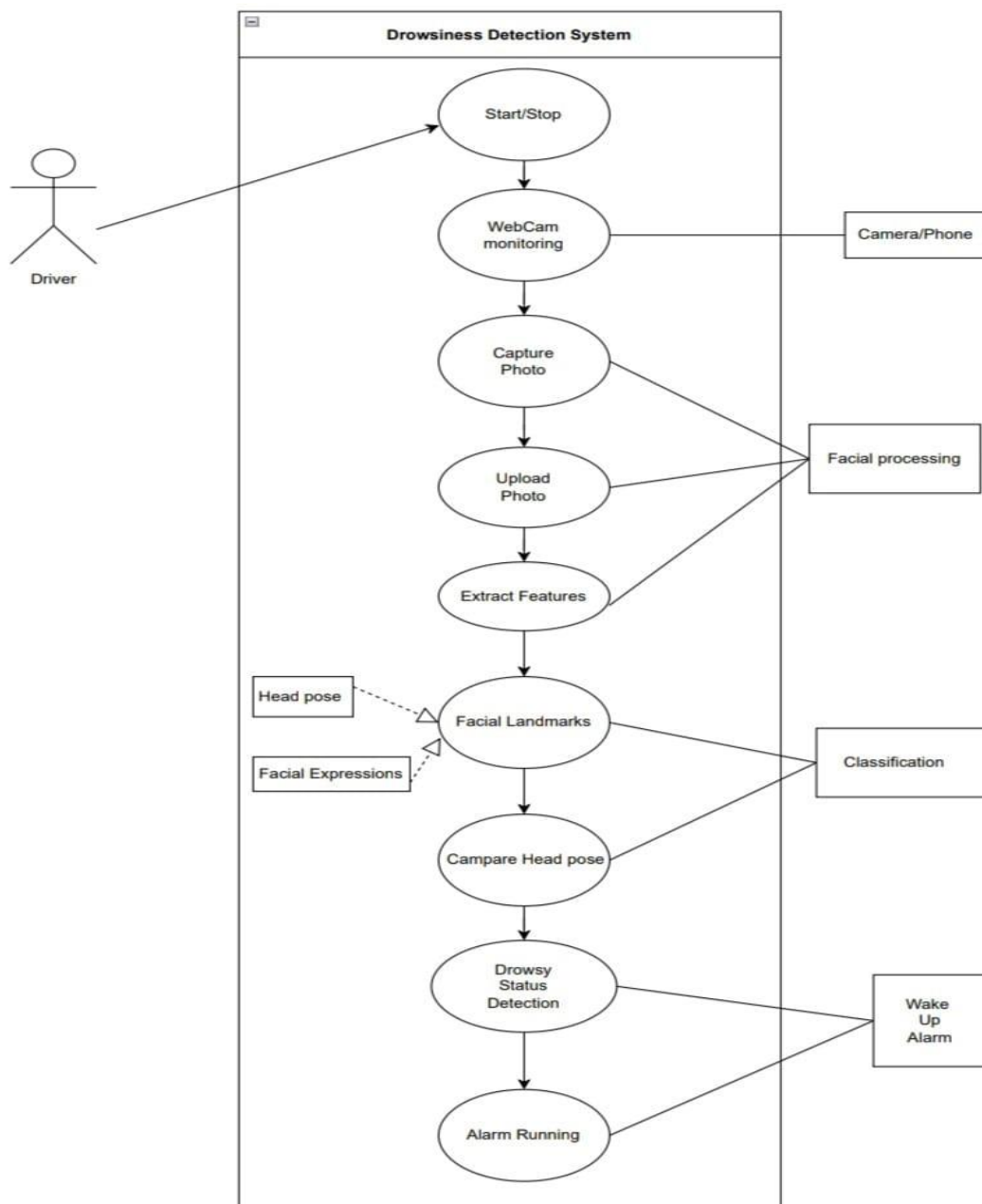
The use case diagram consists of **actors, use cases and relationships**. Actors are external entities that interact with the system. These can include users, other systems, or hardware devices. In the context of a Use Case Diagram, actors initiate use cases and receive the outcomes.

The relations present in use case diagram are **Include, Extend, Association, Dependency, Generalization** relationships.

Notations in Use Case Diagram:



Driver Drowsiness System Use Case Diagram:



2. Class Diagram

Class diagram is a static diagram. It represents the static view of an application. Class diagram is not only used for visualizing, describing, and documenting different aspects of a system but also for constructing executable code of the software application. Class diagram describes the attributes and operations of a class and also the constraints imposed on the system. The class diagrams are widely used in the modelling of object oriented systems because they are the only UML diagrams, which can be mapped directly with object-oriented languages. Class diagram shows a collection of classes, interfaces, associations, collaborations, and constraints. It is also known as a structural diagram.

Notation of Class Diagram:

A class notation consists of three parts:

1. Class Name

- The name of the class appears in the first partition.

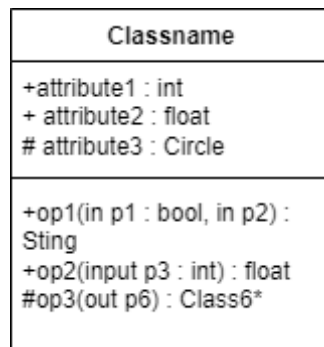
2. Class Attributes

- Attributes are shown in the second partition.
- The attribute type is shown after the colon.
- Attributes map onto member variables (data members) in code.

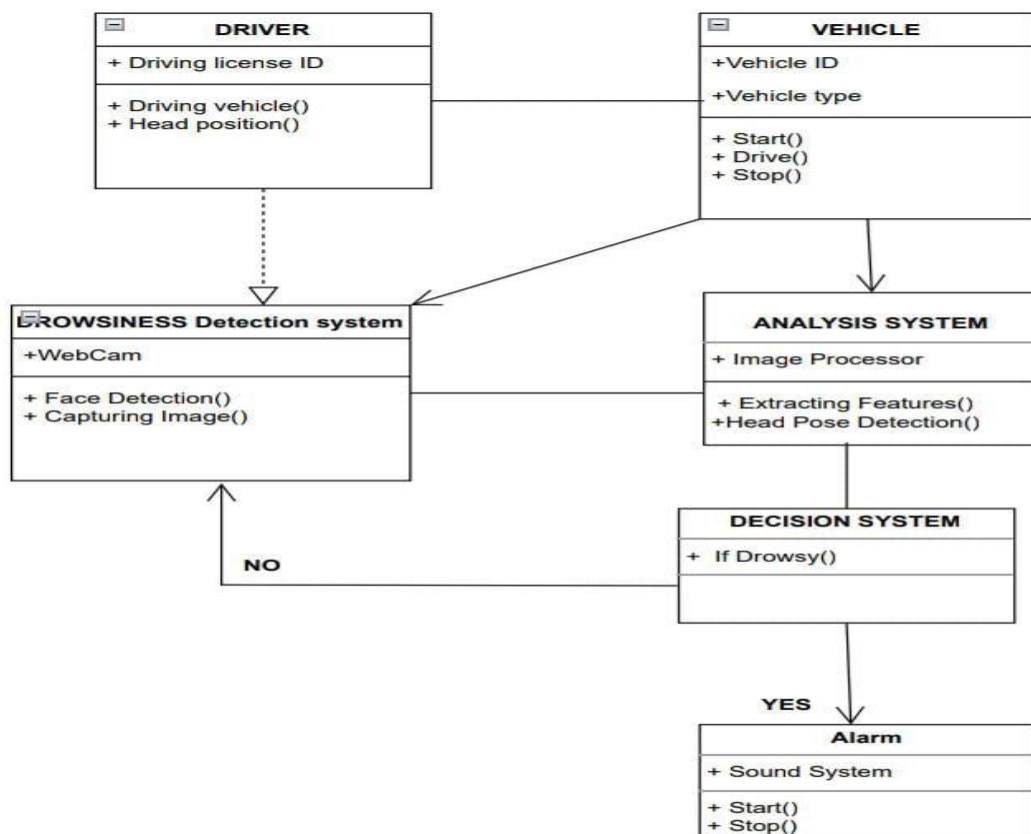
3. Class Operations (Methods)

- Operations are shown in the third partition. They are services the class provides.
- The return type of a method is shown after the colon at the end of the method signature.

- The return type of method parameters is shown after the colon following the parameter name.



Driver Drowsiness Detection Class Diagram:



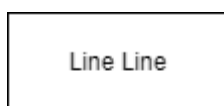
3. Sequence Diagram

The sequence diagram represents the flow of messages in the system and is also termed as an event diagram. It helps in envisioning several dynamic scenarios. It portrays the communication between any two lifelines as a time-ordered sequence of events, such that these lifelines took part at the run time. In UML, the lifeline is represented by a vertical bar, whereas the message flow is represented by a vertical dotted line that extends across the bottom of the page. It incorporates the iterations as well as branching.

Purpose of a Sequence Diagram:

1. To model high-level interaction among active objects within a system.
2. To model interaction among objects inside a collaboration realizing a use case.
3. It either model generic interactions or some certain instances of interaction.

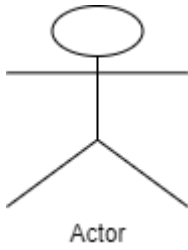
Notations of Sequence Diagram:



LIFE LINE

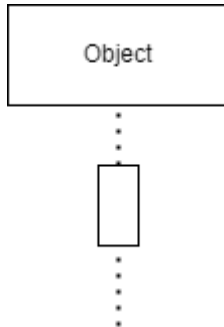


An individual participant in the sequence diagram is represented by a lifeline. It is positioned at the top of the diagram.



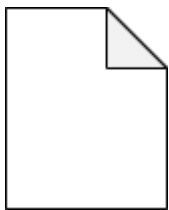
ACTOR

A role played by an entity that interacts with the subject is called as an actor. It is out of the scope of the system. It represents the role, which involves human users and external hardware or subjects.



ACTIVATION

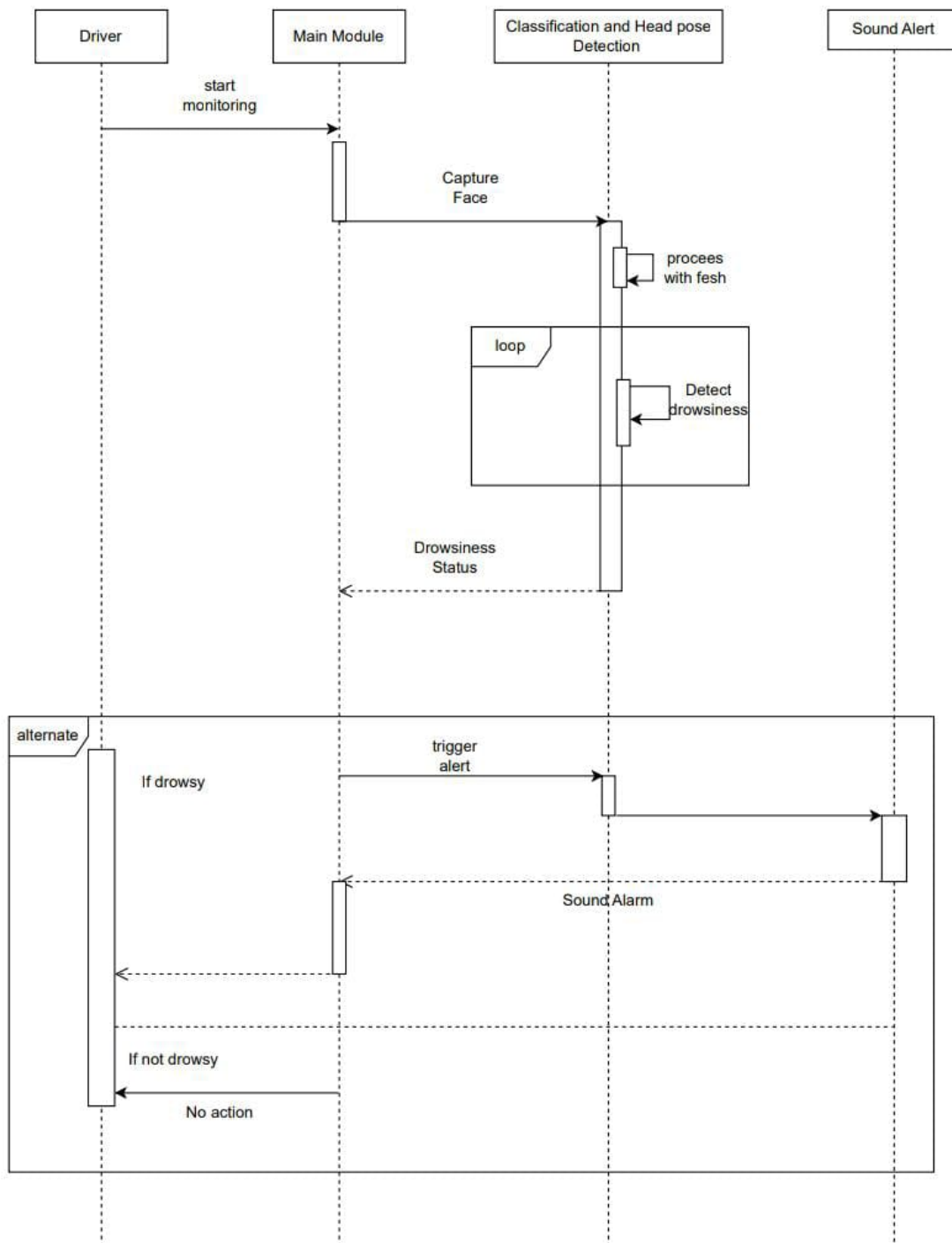
It is represented by a thin rectangle on the lifeline. It describes that time period in which an operation is performed by an element, such that the top and the bottom of the rectangle is associated with the initiation and the completion time, each respectively.



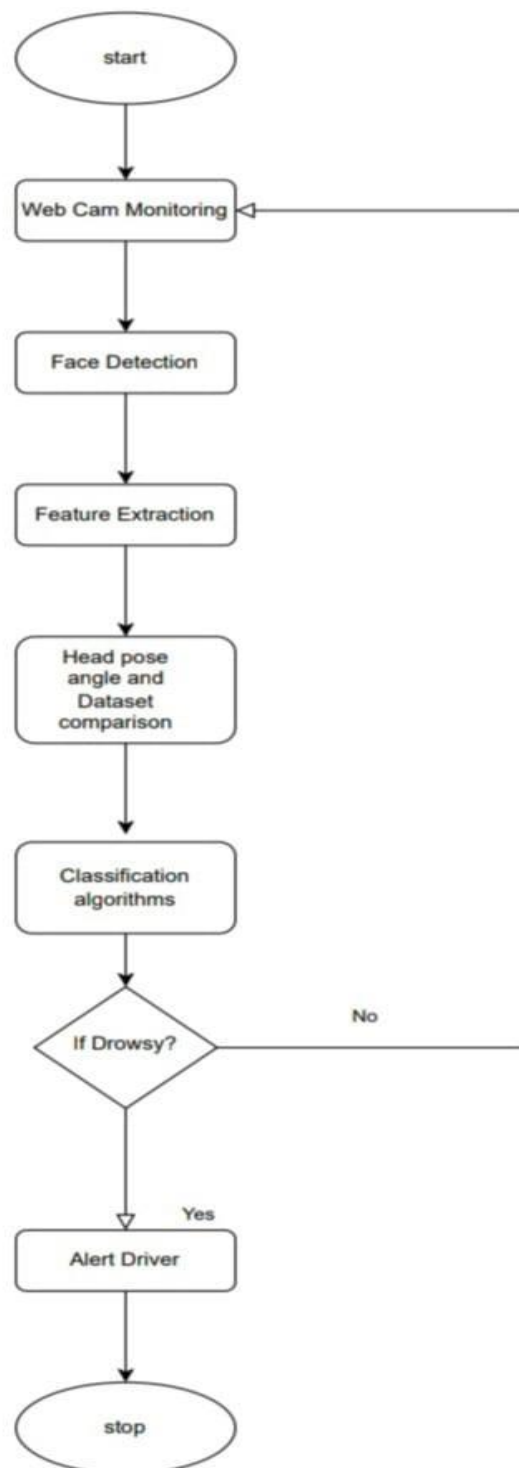
NOTE

A note is the capability of attaching several remarks to the element. It basically carries useful information for the modelers.

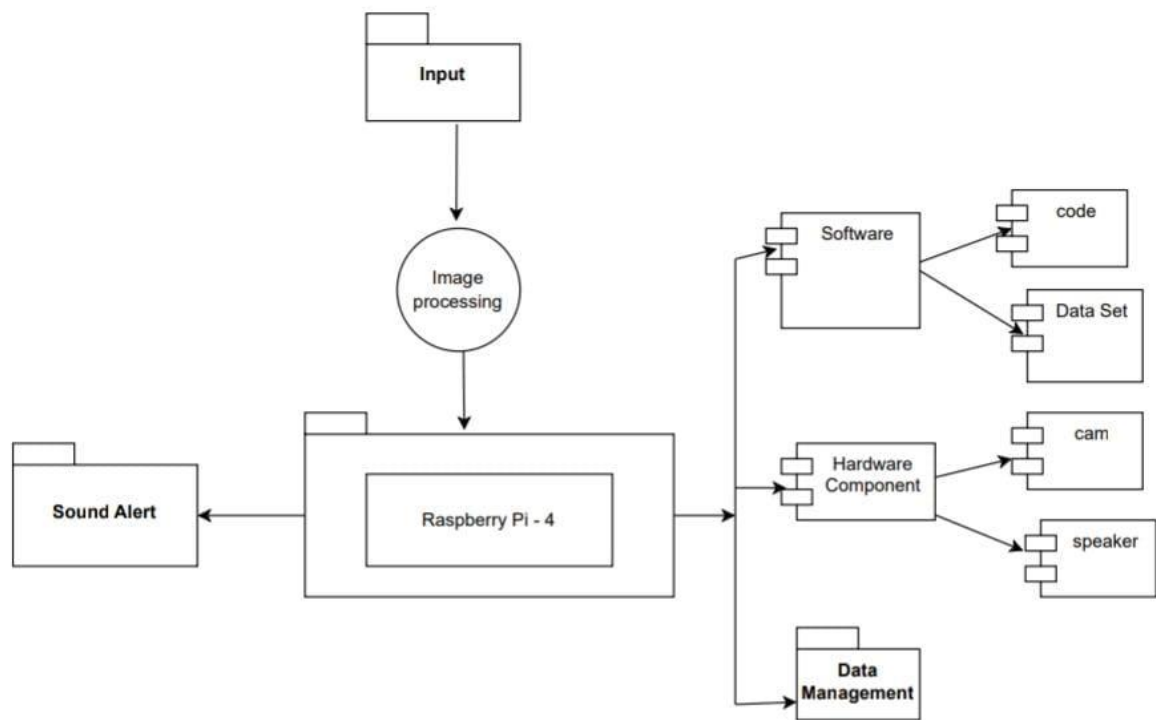
Driver Drowsiness detection Sequence Diagram:



4. ACTIVITY DIAGRAM



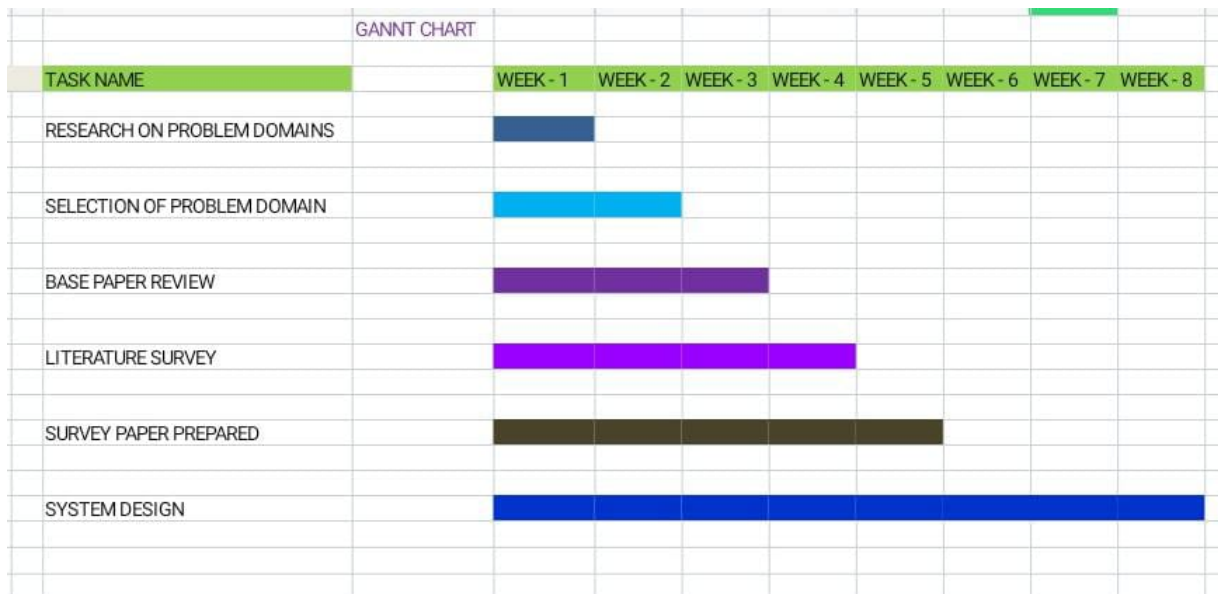
5. PACKAGE DIAGRAM



CHAPTER – 3

Gantt Chart:

Creating a Gantt chart for an integrated Driver Drowsiness Detection Through Face Mesh Solution Using Machine Learning involves breaking down the project into tasks and scheduling them over time. Here is a simplified example of a Gantt chart for such a project.



CHAPTER-4

Conclusion:

The "Driver Drowsiness Detection System Based on Face Mesh Machine Learning Solution" involves, Raspberry Pi Module is programmed with Face Mesh and connected with the camera, and generate alarm when driver is in drowsy state with help of connecting wires. Which can be used as alert mechanism for drowsiness detection.

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